

Glycogen Engineering Improves the Starvation Resistance of Mesenchymal stem cells and their Therapeutic Efficacy in Pulmonary Fibrosis

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This **important** study presents a novel approach to enhance the therapeutic potential of mesenchymal stromal cells (MSCs) by genetically modifying their glycogen synthesis pathway, resulting in increased glycogen accumulation and improved cell survival under starvation conditions, particularly in the context of experimental pulmonary fibrosis. The methods and findings are generally **solid** and could be strengthened in the future by investigating the kinetics of persistence, the immunomodulatory effects, and the underlying improved mechanism of action of MSCs in this pulmonary fibrosis model. If confirmed, this approach could suggest potential methods to improve the therapeutic functionality of MSCs in cell therapy strategies.

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Abstract

Mesenchymal stem cells (MSCs) are widely used in regenerative medicine, including the treatment of pulmonary fibrosis. However, implanted MSCs disappear within days, constraining therapeutic efficacy, which is largely attributed to nutrient deprivation. In this study, we established glycogen metabolism engineering strategies in mammalian cells. By expressing a functionally optimized glycogen synthase (GYSmut), MSCs could accumulate large amounts of glycogen rapidly as a reserve substance. Glycogen engineering significantly improved the survival of MSCs during starvation both in vitro and in vivo, enhancing cell viability post-implantation and their therapeutic efficacy in pulmonary fibrosis. Glycogen-engineered MSCs may serve as chassis cells for further applications. Our research highlights

the importance of glucose metabolism regulation in cell-based therapy and demonstrates the great potential for the metabolic engineering of MSCs and other therapeutic cells.

Introduction

Pulmonary fibrosis (PF) is traditionally regarded as an irreversible lung disease for which there are generally no effective drugs, with the exception of nintedanib and pirfenidone ^{1,2}. Regenerative medicine represents an alternative approach, with several successful therapeutic applications in patients with chronic lung diseases, including PF ^{3–5}. The recent focus on mesenchymal stem cells (MSCs) has generated enthusiasm due to their unique regenerative and immunomodulating properties, including the ability to reverse fibrosis ^{6,7}. Recent studies have begun to unravel the mechanisms underlying the therapeutic potential of MSCs, with a particular emphasis on their secretory properties. The paracrine factors secreted by MSCs, including fibroblast growth factor and cytokines, have been shown to mediate tissue repair and regeneration ^{6,8}. However, the therapeutic efficacy of MSCs has been limited due to their poor survival and short dwell time in the hostile pathological microenvironment, which has hindered the translation of these promising preclinical findings into clinical practice ^{9,10}.

Metabolic regulation is crucial for cell survival post-implantation. It has been reported that the hypoxic in vivo microenvironment inhibits oxidative phosphorylation (OXPHOS) in implanted MSCs, forcing them to rely on glycolysis instead, which yields less ATP ^{11–13}. Our previous single-cell transcriptomic study also showed that the glucose metabolism pathway of MSCs was downregulated in the PF microenvironment ⁶. Compared with in vitro culture systems, it is more difficult for implanted MSCs to obtain sufficient glucose in pathological microenvironments, and recent findings suggest that glucose exhaustion is the main reason for cell death after implantation ¹⁴. Supplying glucose to MSCs improved their survival post-implantation ¹⁵. Therefore, glucose metabolism engineering has the potential to improve MSC therapy by enhancing starvation resistance and prolonging the residence time of implanted cells.

Glycogen is a branched polymer that acts as the primary glucose storage form in mammalian cells, supplying energy under glucose deficiency and promoting cell survival. Glycogen metabolism is controlled by a series of enzymes, including glycogen synthase (GYS) ¹⁶. To reduce cell death of implanted MSCs caused by nutrient deprivation, we engineered the glycogen metabolism of MSCs with genes encoding essential enzymes, which promoted glycogen accumulation (**Fig. 1A**). The accumulated glycogen serves as an energy supply post-implantation, improving the viability and therapeutic efficacy of MSCs (**Fig. 1B**).

Results

Construction of glycogen strategies in mammalian cells

Glycogen is synthesized from UDP-glucose in a process involving glycogenin (GYG), UDP-glucose pyrophosphorylase (UGP), glycogen synthase (GYS), and glycogen-branching enzyme (GBE) ¹⁶. GYG is the initiator of glycogen synthesis, UGP catalyzes the synthesis of UDP-glucose, GYS is responsible for the synthesis of glycogen, and GBE introduces branches into the structure of glycogen (**Fig. 2A**). Here, we firstly devised glycogen engineering strategies based on these essential enzymes.

Initially, we transiently expressed *GYS1* in human HEK293T cells, but it only had a limited, albeit significant effect on glycogen accumulation (**Fig. 2B**). We attributed the unsatisfactory results to substrate limitation, as well as potential inhibition of *GYS1* by cellular pathways. Next, we tried

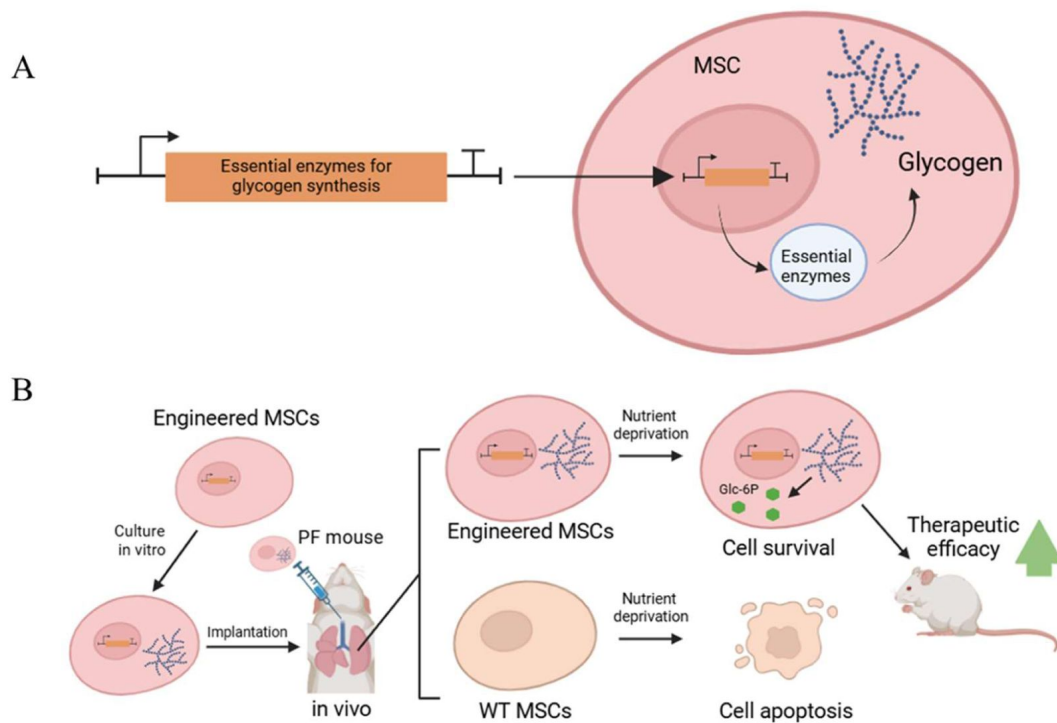


Fig. 1.

Schematic illustration of the strategy used to enhance the therapeutic efficacy of MSCs through glycogen engineering.

(A) Engineering glycogen metabolism of MSCs by overexpressing essential enzymes of glycogen synthesis. (B) Engineered MSCs use glycogen as an energy supply after implantation, improving cell viability and therapeutic efficacy.

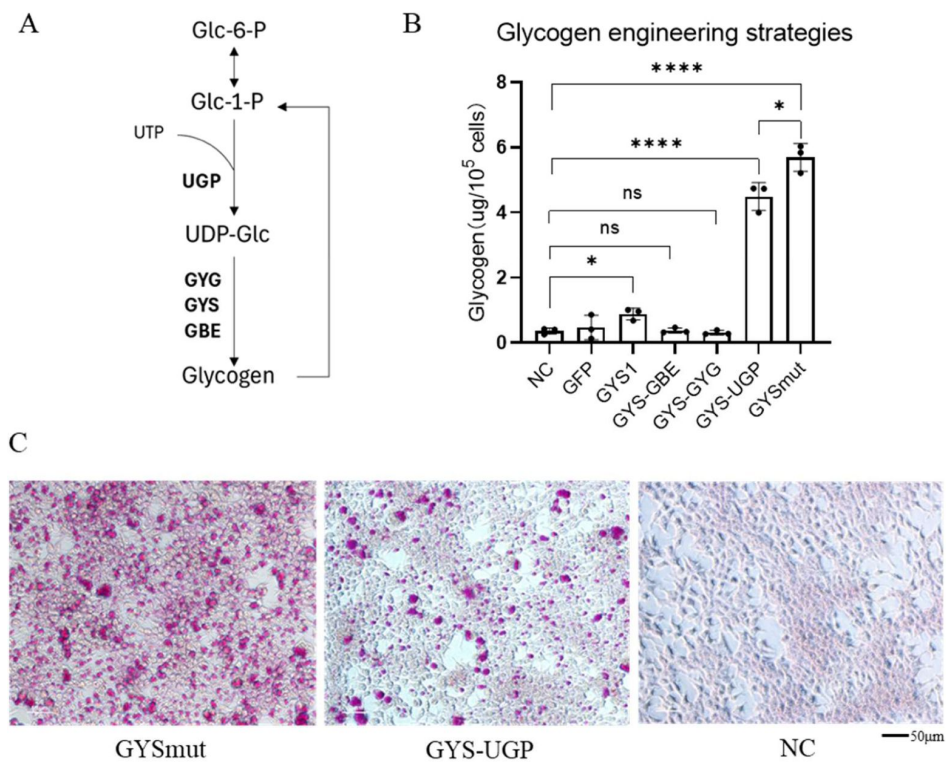


Fig. 2.

Construction of glycogen engineering strategies.

(A) Essential enzymes of glycogen synthesis. (B) Tests of essential enzyme combination strategies in HEK293T cells by transient transfection. The glycogen content of each group was measured. (C) Periodic acid-Schiff (PAS) staining of cells expressing GYSmut and GYS-UGP revealed significant accumulation of glycogen granules.

different expression strategies to increase glycogen accumulation, including (1) expressing *GYS1* alone to promote glycogen synthesis, (2) co-expressing *GYS1* with *GBE1* to increase glycogen-branching, (3) co-expressing *GYS1* with *GYG1* to increase glycogen initiation, and (4) co-expressing *GYS1* with *UGP1* to increase the substrate supply. Some protein kinases (e.g., AMPK) inactivate *GYS1* by phosphorylation at specific serine sites¹⁷. Therefore, (5) we introduced Ser-Ala mutations at the Ser-8, Ser-641, Ser-645, Ser-649, Ser-653, Ser-654, and Ser-657 sites of *GYS1* (GYSmut) to prevent phosphorylation. Both co-expression of *GYS1* with *UGP1* and expression of GYSmut alone resulted in greatly increased glycogen accumulation compared to the control (~12 and 16 folds) and other strategies (Fig. 2B). Glycogen granules were clearly visible following periodic acid-Schiff (PAS) staining (Fig. 2C).

Since most of the Ser-Ala mutation sites were located near the C-terminus of GYSmut, we truncated the 100 C-terminal amino acids to generate GYSdelc. Although this truncated variant was still capable of inducing cellular glycogen accumulation, it exhibited a reduced rate in comparison to full-length GYSmut (Fig. S1A). Further, we co-expressed GYSmut with UGP, which demonstrated a marked additive effect (Fig. S1B).

Glycogen accumulation improved the starvation resistance of MSCs

After expressing GYSmut in MSCs by lentivirus transduction, glycogen accumulation was significantly increased in the engineered MSCs (Fig. 3A). PAS staining showed more PAS-positive glycogen granules in GYSmut MSCs (Fig. 3B), which altered cell morphology by significantly increasing the cell volume, as assessed by flow cytometry (Fig. S2A). Glycogen granules were located both in the cytoplasm and nucleus (Fig. S2B). Interestingly, when co-overexpressing of wild-type *GYS1* and *GYG*, we observed a diffuse intracellular glycogen distribution rather than bulky glycogen particles, indicating that the combination of distinct enzymes influences the characteristics of glycogen synthesis (Fig. S2C).

To evaluate the starvation resistance of glycogen-engineered MSCs, we cultured GYSmut MSCs in Dulbecco's Phosphate-Buffered Saline (DPBS) instead of a complete culture medium to create a nutrient-deprived environment. The Cell-Counting-Kit-8 assay was used to measure cell viability at various time points. Notably, approximately 40% of the GYSmut MSCs were still viable after 72 hours of starvation culture, whereas wild-type MSCs died rapidly after 8 hours (Fig. 3C). After 48 hours of starvation, GYSmut MSCs still retained a 39% glycogen reserve (Fig. 3D), indicating that the accumulated glycogen could be a long-term energy resource to maintain higher cell viability during nutrient deprivation following implantation. Since some pathological contexts are hypoxic, we tested the starvation resistance of MSCs under hypoxia (oxygen concentration < 5%), with the engineered MSCs still maintaining a higher survival proportion (Fig. S2D).

As glycogen is a crucial node of carbon metabolism in mammalian cells, we next investigated whether glycogen engineering caused global changes in MSCs. Cell counting showed that glycogen engineering had no significant effect on cell proliferation (Fig. 3E). To evaluate the effects of glycogen engineering on the adipogenic differentiation capacity of MSCs, we induced GYSmut MSCs with adipocyte medium for 5 days, followed by oil red O staining and qRT-PCR of the adipocyte differentiation marker *Lpl*. The results indicated a decrease in the adipocyte differentiation potential of GYSmut MSCs (Fig. 3F, 3G). It was reported that adipogenic differentiation and the immunomodulatory functions of MSCs are mutually exclusive¹⁸, so blocking adipocyte differentiation may enhance the immunosuppressive activity of MSCs.

We next performed bulk RNA sequencing (RNA-seq) to understand the effect of glycogen engineering on the transcriptome of GYSmut MSCs. A total of 759 genes were up- and 542 were downregulated in GYSmut MSCs, indicating a broad effect on the transcriptome (Fig. S2E). Gene ontology (GO) enrichment analysis revealed that differentially transcribed genes were

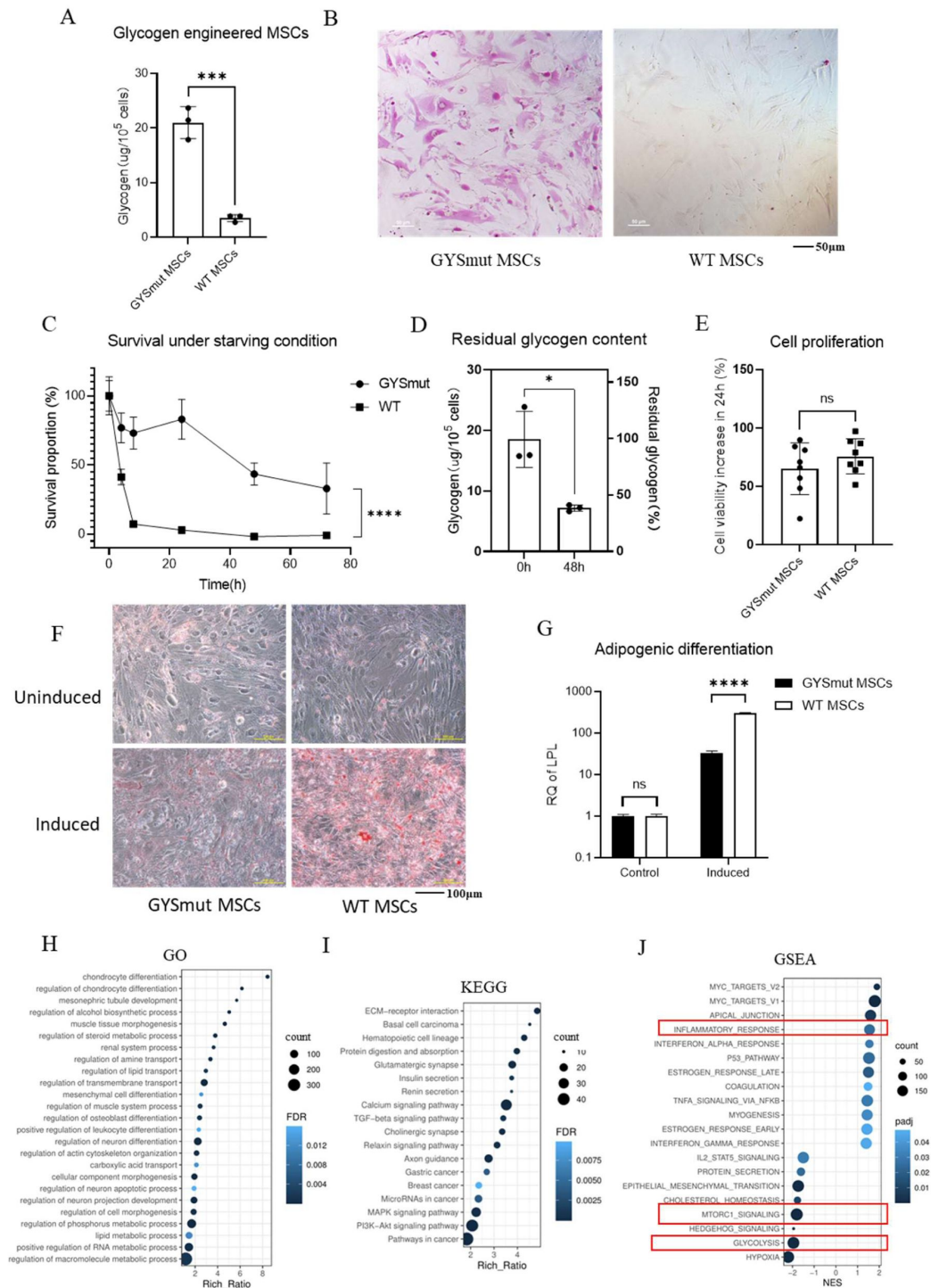


Fig. 3.

Construction of glycogen-engineered MSCs.

(A) Glycogen content of GYSmut MSCs. (B) PAS staining of GYSmut MSCs, showing glycogen granules (red). (C) Survival of engineered MSCs under DPBS (starvation) treatment in vitro (N = 8). (D) Residual glycogen content of GYSmut MSCs after DPBS (starvation) treatment for 48 hours. (E) Viability of GYSmut MSCs according to the CCK8 assay. (F, G) Adipogenic differentiation potential of GYSmut MSCs, assessed by oil red O staining and qPCR detection of Lpl expression (n=3, unpaired *t*-test P-value < 0.0001). (H) GO enrichment analysis of differentially expressed genes (DEGs) between GYSmut MSCs and the GFP control. (I) KEGG analysis of DEGs. (J) Gene set enrichment analysis (GSEA) of the DEGs.

predominantly enriched in processes related to cellular differentiation, morphogenesis, and the regulation of various metabolic pathways, including lipid metabolism, reflecting potential metabolic reprogramming. The differentially expressed genes were also enriched in the terms related to processes of muscle, kidneys, and neural tissue, which are sites where glycogen is naturally distributed in mammals (**Fig. 3H** [↗](#)) ¹⁶[↗](#). KEGG analysis showed the PI3K-Akt signaling pathway was also modulated, which is crucial for cell survival (**Fig. 3I** [↗](#)). The GSEA results showed that the inflammatory response gene set was upregulated, which may be beneficial in the treatment of PF, while mTORC1 and glycolysis-related gene sets were downregulated (**Fig. 3J** [↗](#)). It was reported that mTORC1 responds to cellular amino acid and glucose levels ¹⁹[↗](#), which is in agreement with the hypothesis that glycogen synthesis affects cellular glucose levels and the process of glycolysis.

Glycogen engineering promoted the survival of MSCs post-implantation

In our previous study, intratracheal administration of MSCs exhibited a robust therapeutic effect in a mouse model of bleomycin-induced pulmonary fibrosis (PF) ⁶[↗](#), and further scRNA-seq analysis of implanted MSCs within the PF microenvironment revealed downregulation of glucose metabolism (Fig. S3), suggesting that nutrient availability is a critical determinant of MSC viability and may potentially limit their therapeutic efficacy. We, therefore, tested whether glycogen engineering would promote the survival of GYSmut MSCs *in vivo*. PF mice were intratracheally injected with glycogen-engineered MSCs co-expressing GYSmut and *Gaussia* luciferase (GYSmut-Gluc), while the control group was injected with Gluc MSCs. We collected lungs of PF mice on days 0 and 7 after MSC implantation, and measured the fluorescence intensity of lung tissue homogenates with a commercial kit, to assess the counts of living MSCs (3 mice each group each time point) (**Fig. 4A** [↗](#)). A higher proportion of surviving GYSmut-Gluc MSCs was detected than in the control (**Fig. 4B** [↗](#)).

Because of the limited sensitivity of *Gaussia* luciferase labeling for *in vivo* imaging, to assess the persistence and biodistribution of implanted MSCs within the PF microenvironment, we used a different luciferase, *Akaluc* luciferase. GYSmut MSCs co-expressing *Akaluc* (GYSmut-Akaluc) were intratracheally injected into PF mice. This approach, in combination with the substrate Akalumine ²⁰[↗](#), allows for highly sensitive live imaging and thus provides insights into the retention of implanted cells. In the control group, MSCs expressing *Akaluc* were injected (5 mice each group). As expected, on day 7 post-implantation, higher fluorescence intensity was detected in the GYSmut MSCs group compared with the control (**Fig. 4C** [↗](#), Fig. S4), indicating the greater survival ability of GYSmut MSCs compared to wild-type MSCs in the PF microenvironment. Quantification of fluorescent imaging revealed a further decline in survival rates to similar levels in both GYSmut MSCs and the control by day 11, suggesting that glycogen engineering appears to be primarily functional during the first two weeks post-implantation (Fig. S4).

Engineered MSCs demonstrated a beneficial effect on pulmonary fibrosis

To investigate the therapeutic efficacy of glycogen-engineered MSCs (GYSmut MSCs), we administered them via tracheal injection into mice on day 3 following bleomycin treatment, and recorded the survival rate and body weight of mice over time (10 mice each group). GYSmut MSCs demonstrated a more pronounced effect than the control group expressing only GFP (**Fig. 5A** [↗](#)). Quantitative analysis of lung tissue sections stained with H&E and Masson's trichrome revealed that the GYSmut MSCs group exhibited superior therapeutic efficacy on day 7, as evidenced by reduced collagen deposition and preserved alveolar size (quantified by mean linear intercept, MLI) compared to the control (**Figs. 5B and C** [↗](#)). The improved therapeutic efficacy illustrated the beneficial effect of enhanced MSC viability and confirmed the rationale for glycogen engineering.

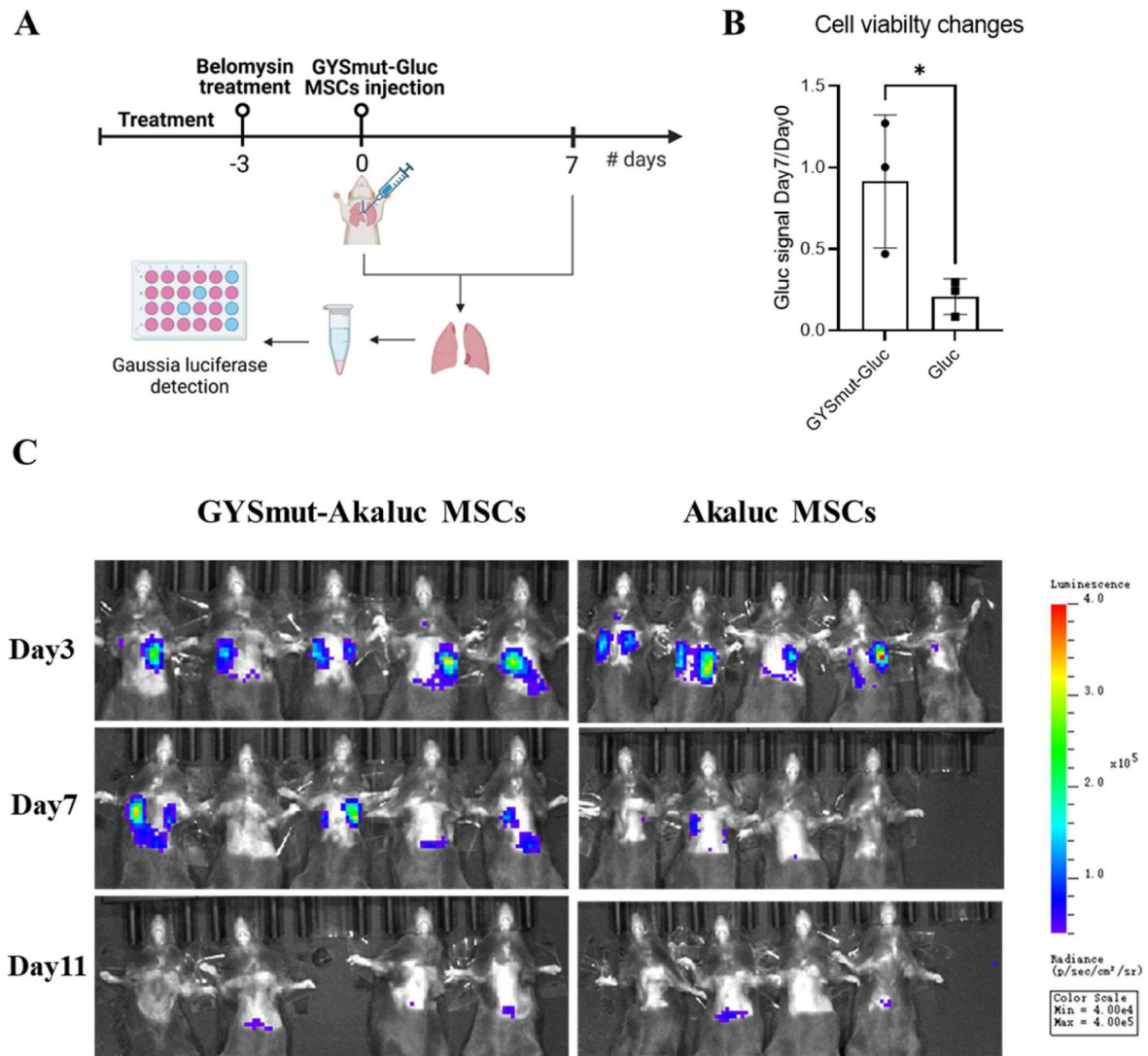


Fig. 4.

Survival of implanted glycogen-engineered MSCs.

(A) Schematic illustration of the strategy used to assess the survival of MSCs by detecting *Gaussia* luciferase activity in the homogenate of lungs. (B) Changes of *Gaussia* luciferase activity in the two groups on day 7 post-implantation (3 mice each group, unpaired *t*-test *P*-value = 0.044). (C) Live imaging of *Akaluc* luciferase activity in the two groups implanted with GYSmut-Akaluc MSCs and control cells (5 mice each group). One mouse from the control group died on day 7, and one mouse from the GYSmut-Akaluc group died on day 11.

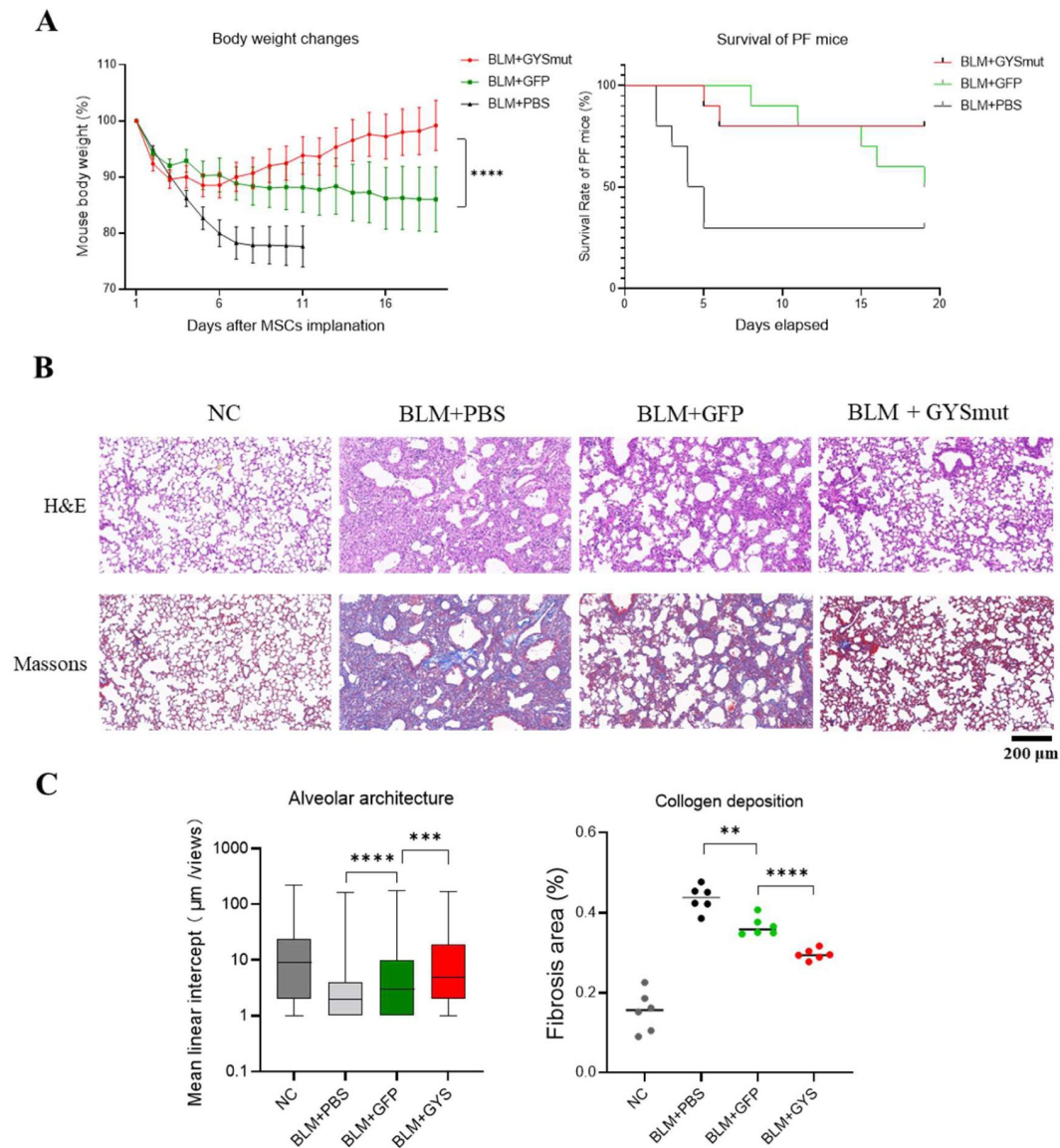


Fig. 5.

Therapeutic efficacy of glycogen-engineered MSCs.

(A) Survival and body weight changes of PF mice treated with GYSmut MSCs and control cells (10 mice each group, mean $\pm$ SEM, 2-way ANOVA P-value=0.0001), (B) Representative lung tissue sections stained with H&E and Masson's trichrome. NC group is healthy mice. (C) Collagen deposition and preserved alveolar size (quantified by mean linear intercept, MLI) of lung tissue sections (6 mice each group).

Discussion

Glycogen is found in a wide range of organisms, ranging from bacteria to mammals. As a major form of stored glucose, glycogen plays an essential role in maintaining energy homeostasis and cell survival. In this study, we engineered the glycogen metabolism of MSCs to promote glycogen accumulation. We tested several engineering strategies in human HEK 293T cells and primary MSCs derived from mice. Over-expression of phosphorylation-resistant GYS (GYSmut), as well as the co-expression of GYS and UGP, induced intense glycogen. Glycogen engineering enhanced the starvation resistance of MSCs and improved their viability post-implantation. Compared with the control group, the engineered MSCs exhibited a significantly enhanced therapeutic effect on pulmonary fibrosis, indicating the importance of glucose metabolism regulation for MSC-based therapy.

Glycogen-engineered MSCs may serve as chassis cells for further applications by enhancing cell viability post-implantation, and our glycogen engineering strategies also have the potential to be applied to other therapeutic cells. Transient expression of GYSmut by mRNA transfection may be a safer alternative than lentiviral transduction, avoiding the risks of chromosomal modification. LNP-mediated circular mRNA transfection would further improve engineering efficiency²¹.

We investigated the effect of glycogen engineering on the transcriptome of MSCs by RNA-seq, which revealed large-scale gene transcription changes. In future studies, complex regulatory networks may be elucidated at the levels of proteomics and metabolomics. Previous studies have revealed roles of glucose metabolism regulation in MSCs' immunoregulatory properties^{22, 23}. The broader impact of glycogen engineering on MSCs remains to be elucidated. We tested the therapeutic efficacy of glycogen-engineered MSCs in the bleomycin-induced PF mice model. Their efficacy in other disease models is still to be investigated.

Within the context of MSC-based cellular therapies, administered cells were distributed to various microenvironments characterized by variable oxygen concentration, nutrient availability, and immune responses. While *in vivo* animal models offer a more physiologically relevant platform for validation, they are less convenient for detailed investigation due to multifactorial influences, making it challenging to fully understand the precise kinetic properties and adaptation dynamics of the implanted cells, which is a limitation of this study.

In previous studies, complex gene circuits were introduced to enhance the efficacy of various cell-based therapies, including MSCs and CAR-T cells^{24, 25}, and some research also revealed the advantages of metabolic engineering (ME)²⁶. However, in contrast to extensive ME research in prokaryotic organisms, ME of mammalian therapeutic cells has yet to be developed. Here, we demonstrated the great potential of glycogen engineering, offering a basis for future studies. By dynamically controlling essential enzymes, cellular metabolism can be remodeled to achieve optimal survival and therapeutic efficacy post-implantation. Further research in this area will provide additional opportunities for cell therapy, gene therapy, and tissue engineering.

Materials and methods

Animals and experimental design

All the animal experiments were authorized by the institutional animal care and use committee of Tsinghua University. Six-week-old male C57BL/6J mice were housed in the specific pathogen-free experimental animal environment at the laboratory animal research center of Tsinghua

University, with 5~6 mice per cage having access to sterilized food and water ad libitum under a 12-hour light/dark cycle. Mice were anesthetized with avertin (200 μ L per mouse) or 2.5% tribromoethanol. And euthanized by cervical dislocation or CO₂ inhalation. 76 mice were used in total. To obtain reliable results, each group contains at least 3 mice. In the same experiment, the mice in each group came from the same batch and were randomly captured, processed and assigned to different cages. The number of mice is randomly assigned, the surgeon knows the grouping situation, and the statistical analyst does not know the grouping in advance. All animal experiments were authorized by the institutional animal care and use committee of Tsinghua University (Approval No. 23-WQ1.G23-1, Study on different factors on the survivability of implanted MSCs in murine pulmonary fibrosis model, 2023-01-18 to 2026-04-12).

Isolation of mouse adipose-derived stem cells

MSCs were obtained by euthanizing the mice, digesting the inguinal adipose tissue with collagenase type I (1.0 mg/ml, Sigma), and culturing the released cells in mouse ADSC basal medium (CYAGEN, China) in a humidified atmosphere containing 5% carbon dioxide at 37°C, according to a previously published method [6](#).

Construction of the BLM-induced PF mouse model and administration of MSCs

Mice were anesthetized with 2.5% tribromoethanol (0.8% NaCl, 1 mM Tris (pH 7.4), 0.25 mM EDTA (pH 7.4)) and administered 3.5 mg/kg bleomycin solution (Sigma, B5507-15un) in 50 μ L phosphate-buffered saline (PBS) intratracheally on day 0 using a 1 mL syringe with a 25G needle. Control animals were mock-treated with PBS [6](#).

Injection of MSCs (at passages 3~6) was performed on day 3 after BLM treatment. Each animal received an intratracheal injection of 50 μ L of a suspension comprising 1×10^7 cells/mL in PBS (5×10^5 cells/mouse) using a 1 mL disposable syringe with a 25G needle. For the control and BLM + PBS groups, 50 μ L PBS was injected instead. Before implantation, the cells were washed three times to remove the culture medium. The body weight of treated mice was recorded daily [6](#), [27](#).

Histological evaluation of lung damage and collagen deposition of BLM-induced lung injury

To assess the development of PF and therapeutic efficacy, mice were evaluated on day 7. For lung tissue collection, mice were euthanized by CO₂ inhalation. The thoracic cavity was opened and the lungs were perfused and washed with ice-cold PBS. The lungs were then fixed with 4% (v/v) paraformaldehyde in PBS and processed into 5- μ m-thick paraffin sections, which were mounted on glass slides.

For histological assessment of alveolar architecture and fibrosis, sections were stained with hematoxylin and eosin (H&E) or Masson's trichrome stain, respectively. The samples were scanned using Panoramic SCAN (3DHISTECH) with a 20 $\times$ objective. Fibrosis area percentages and alveolar size (quantified by mean linear intercept, MLI) were calculated according to a previously published method [6](#), [28](#).

Generation of genetically engineered MSCs and in vitro tests

HEK293T cells were cultured in DMEM with 10% FBS. Sequences of essential enzymes are amplified from cDNA of mouse MSCs. To test combinations of essential enzymes, HEK293T cells in 6-well plates were transiently transfected with encoding plasmids (2 μ g/well) using Lipo8000 (Beyotime, C0533). Co-expressed enzymes were connected by the 2A sequence. After 3 days, the glycogen content of each group was quantified using a commercial kit (SOLARBIO, BC0345-100T). A PAS staining kit (Beyotime, C0142S) was used to stain glycogen granules.

To construct the lentiviral vectors, designed gene sequences were cloned into the pCDH backbone and used to co-transfect HEK293T cells with pMD2.G and psPAX2 plasmids using Lipo8000. Essential enzymes were co-expressed by connecting them with EGFP or puromycin N-acetyltransferase via the 2A sequence. The cell supernatant was collected every 24 hours for 3 days. The viral particles were concentrated using Lentivirus Concentration Solution (Servicebio, G1801-100ML), after which the infectious solution was prepared by resuspending them in 200 μ L PBS and stored at -80°C until use. When the density of MSCs in the 6-well plate reached approximately 80%, 20 μ L of the lentiviral solution was added. Following 2 days after infection, puromycin was added to a final concentration of 1 mg/L for selection.

Glycogen was quantified after 3 days of induction. To simulate starvation conditions, DPBS solution was used to replace the culture medium of seeded MSCs (96-well plate, 5000 cells/well), and a CCK-8 assay kit (BIORIGIN, BN15201) was used to measure cell viability. To test the adipogenic differentiation potential of MSCs, we induced them with commercial adipocyte medium (Procell, PD-027) for 5 days as described in the manufacturer's protocol, and then performed oil red O staining (Beyotime, C0157S).

Viability detection and live imaging of MSCs post-implantation in the PF model

To detect viability changes of MSCs post-implantation, 2.5×10^5 MSCs expressing *Gaussia* luciferase suspended in 50 μ L of PBS were injected intratracheally into PF mice, as described above. The whole lungs of mice were collected on the day of surgery (D0) and the 7th day (D7) after surgery (n=3 each time point), and were stored in a 1.5 mL centrifuge tube at -80 °C. Then, 400 μ L PBS and 20 1-mm-beads (Servicebio, G0201) were added to the centrifuge tube with lungs for processing with a KZ-III high-speed homogenizer (Servicebio) at 60Hz for 30s, 6 times. The homogenate was centrifuged (10000g, 5 min) and the activity of *Gaussia* luciferase in the suspension was detected using a commercial kit (GeneCopoeia, LF061). The lungs of cells without *Gaussia* luciferase expression were used as negative controls, and the corresponding signals were considered to represent background subtraction in the calculations. Cell viability changes of each group were assessed by calculating the ratio of the intensity of lung tissue collected on D7 to D0.

To detect the in vivo distribution of MSCs post-implantation, MSCs expressing Akaluc were injected intratracheally into PF mice (n=5), as described above. Live imaging was performed on the 3rd and 7th days after implantation. Akalumine-HCL (InvivoChem, V41343) was dissolved in PBS (2.5 mg/mL) and intraperitoneally injected into mice (100 μ L/mouse). Luminescence was detected after 10 minutes using a Lumina III instrument and analyzed with Living Image software version 4.8.

RNA-seq of MSCs

MSCs were cultured in 100-mm dishes (5×10^6 cells per dish) in DMEM/F-12 medium. Trizol (Vazyme, R401-01) was used for RNA extraction. Illumina RNA-seq and subsequent analysis were performed by ANNOROAD Corporation (Beijing, China). DESeq2 was used to identify differentially expressed genes ($|\log_2 \text{fold-change}| \geq 1$, adjusted p-value < 0.05).

Statistical analysis

Statistical analysis was performed using GraphPad Prism software version XY (GraphPad Software Inc.). Significant differences between groups were identified using unpaired Student's *t*-test or 2-way ANOVA. Asterisks were used to indicate significance in figures (*: p<0.05, **: P<0.01, ***: p<0.001, ****: p<0.0001). Unless indicated otherwise, all quantitative data are presented as mean values $\pm$ SD.

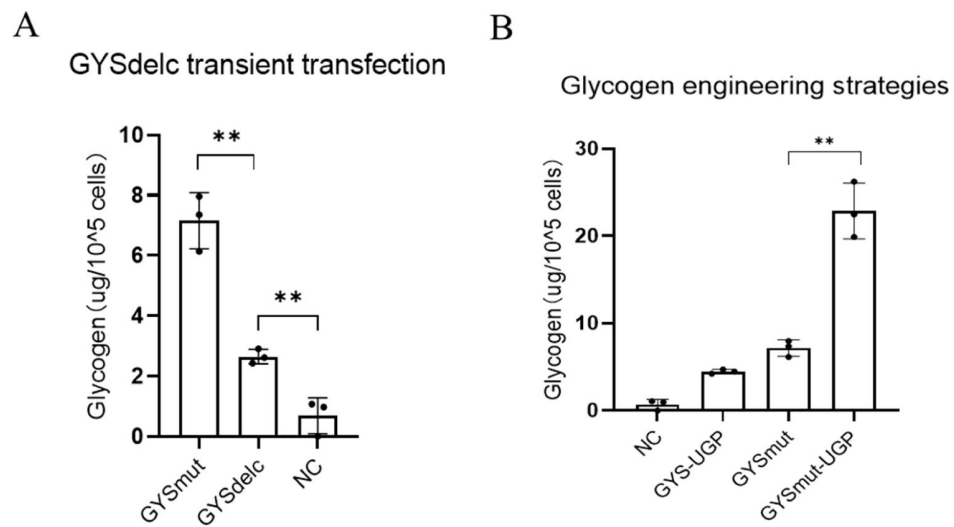


Fig. S1.

(A) GYSdelc promoted glycogen accumulation. (B) Co-expressing GYSmut and UGP further promoted glycogen accumulation.

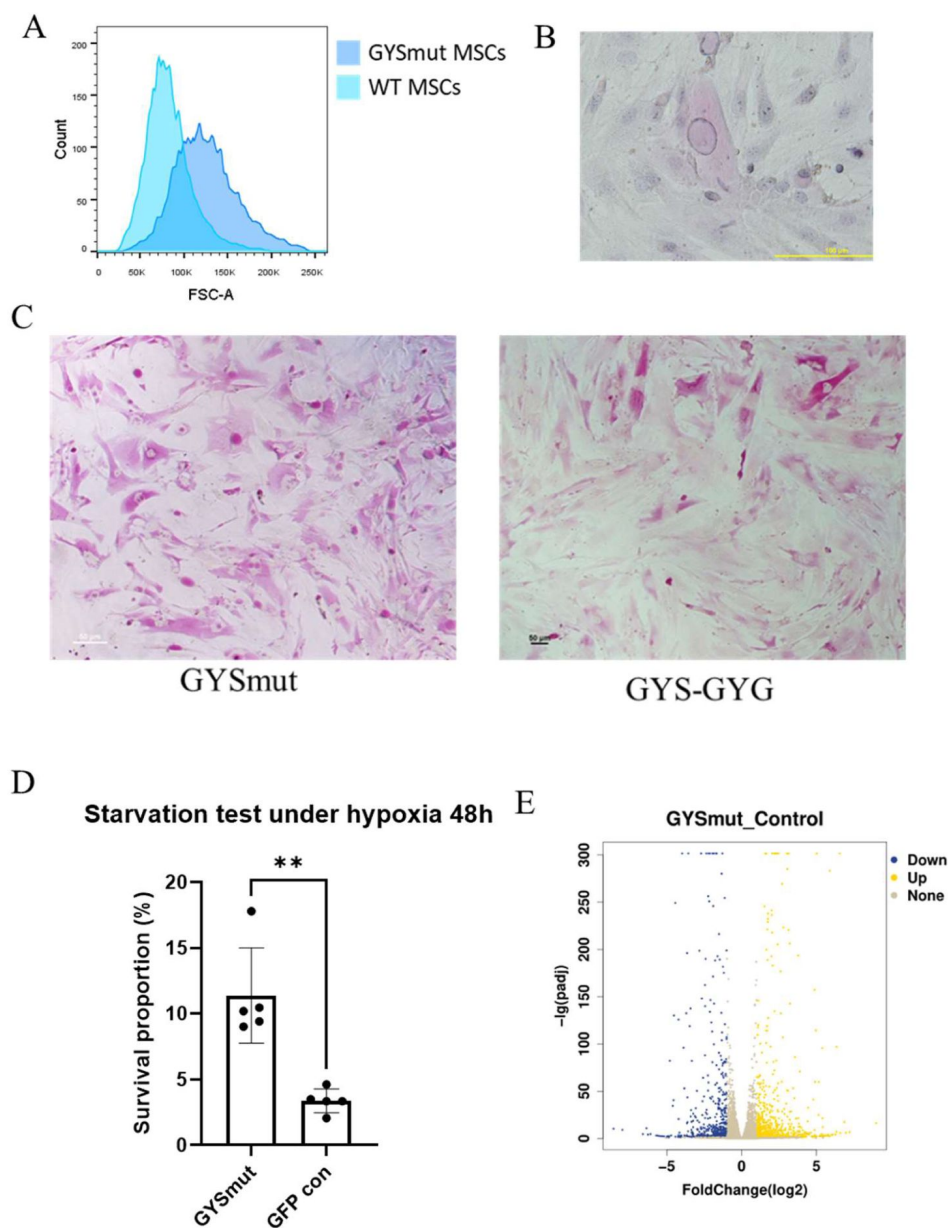


Fig. S2.

Impacts of glycogen engineering on MSCs.

(A) Cell volume change of engineered MSCs, assessed through flow cytometry. (B) PAS and hematoxylin staining of GYSmut MSCs. Glycogen (red) is distributed in cell nucleus (blue) and cytoplasm. (C) Glycogen distribution of GYSmut and GYS-GYG MSCs. Scale bar, 50 μ m. (D) Starvation resistance test under hypoxia. (E) Impacts of glycogen engineering on transcriptome of MSCs.

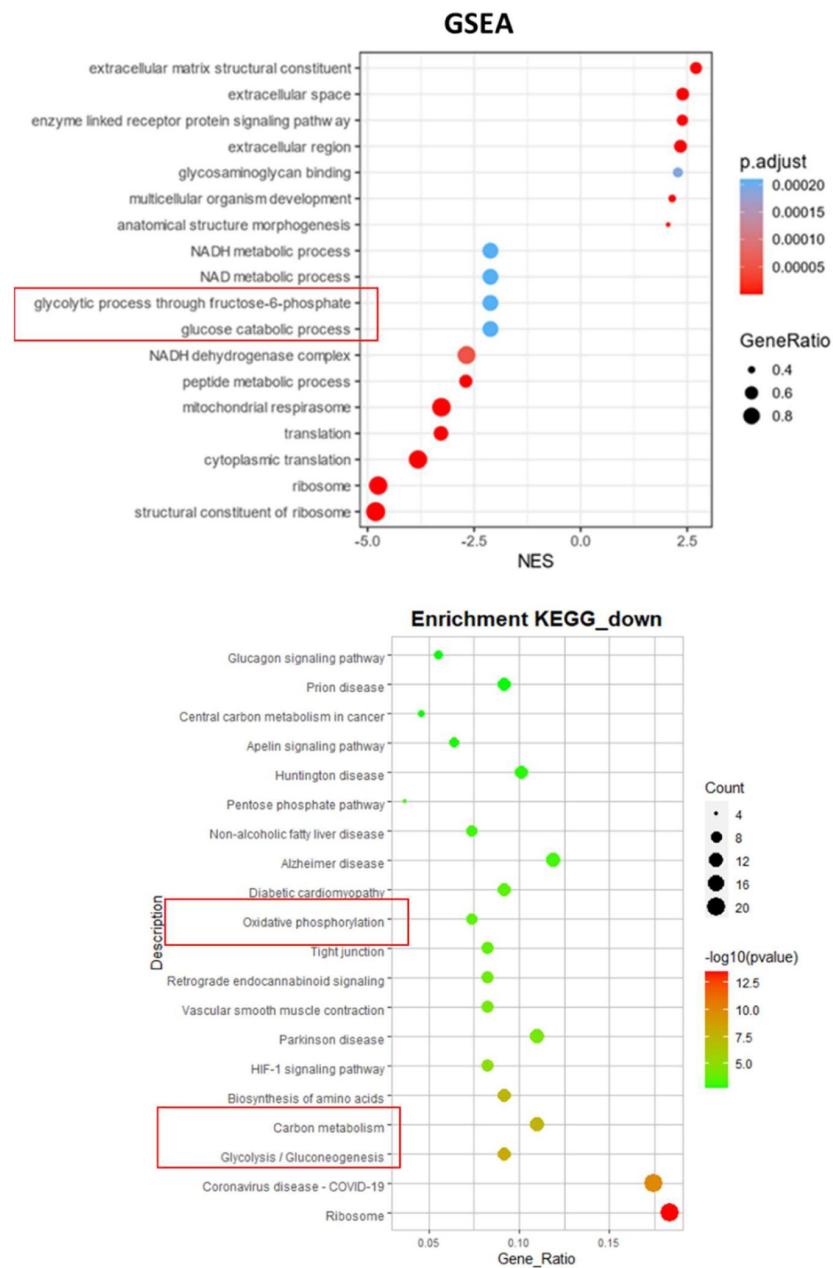


Fig. S3.

KEGG and GSEA analysis of implanted MSCs in our previous research.

GFP MSCs were intratracheally administered to PF mice and collected through flow cytometry. Single-cell sequencing showed downregulation of glucose metabolism.

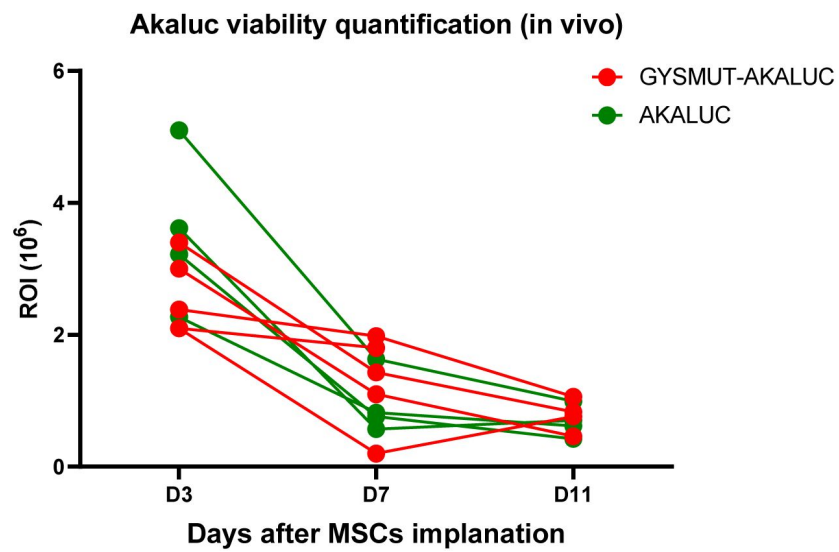


Fig S4

Quantification of akaluc activity of in vivo imaging post-implantation.

GYS1(wt)	MPLSRSLSVSSLPGLEDWEDEFDPENAVLFEVAWEVANKVGGIYTVLQTKAKVT GDEWGDNYLVGPYTEQGVRTQVELLEPPTPELKRTLDSMNSKGCKVYFGRWLI EGGPLVVLDDVGASAWALERWKGELWDTCNIGVPWYDREANDAVLFGFLTTFW LGEFLAQNEEKPYVVAHFHEWLAGVGLCLCRARRLPVATIFTTHATLLGRYLCA GAVDFYNNLENFNVDKEAGERQIYHRYCMERAAAHCAHVFTTVSQITAIEAQHL LKRKPDIVTPNGLNVKKFSAMHEFQNLHAQSKARIQEFVRGHFYGHLDNFNDKTL YFFIAGRYEFSNKGADVLEALARLNYLLRVNGSEQTVVAFFIMPARTNNFNVET LKGQAVRKQLWDTANTVKEKFGRKLYESLLVGSLPDMNKMLDKEDFTMMKRAI FATQRQSFPPVCTHNMLDDSSDPILTTIRRIGLFNSSADRVKVIHFPEFLSSTSPLLP VDYEEFVRGCHLGVFPSYYEPWGYTPAECTVMGIPSISTNLSGFGCFMEEHIADPS AYGIYILDRRFRSLDDSCSQLTSFLYSFCQQSRRQRIIQRNRTERLSDLLDWKYLGR YYMSARHMALAKAFPDHFTYEPHEVDATQGYRYPRPASVPPSPSLSRHSSPHQSE DEEPRDGPLGEDSERYDEEEEAAKDRRNIRAPEWPRRASCSSSTGGSKRSNSVDT GPSSSLSTPTEPLSPTSSLGEERN
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Table 1

Protein sequences used in this study

GYSmut	MPLSRSLAVSSSLPGLEDWEDEFDPENAVLFEVAWEVANKVGGIYTVLQTKAKVT GDEWGDNYL VG PYTEQGVRTQVELLEPPTPELKRTLDSMNSKGCKVYFGRWLI EGGPLVLLDVGASAWALERWKGELWDTNIGVPWYDREANDAVLFGFLTTFW LGEFLAQNEEKPYVVAHFHEWLAGVGLCLCRARRLPVATIFTTHATLLGRYLCA GAVDFYNNLENFNVDKEAGERQIYHRYCMERAAAHCAHVFTTVSQITAIEAQHL LKRKPDIVTPNGLNVKKFSAMHEFQNLHAQSKARIQEFVRGHFYGHLDNFNLDKTL YFFIAGRYEFSNKGADVLEALARLNLLRVNGSEQTVVAFIMPARTNNFNVT LKGQAVRKQLWDTANTVKEKFGRKLYESLLVGSPLDMNKMLDKEDFTMMKRAI FATQRQSFPPVCTHNMLDDSSDPILTIRRIGLFNSSADRVKVFHFPEFLSSTSPLLP VDYEEFVRGCHLGVFPSYEPWGYTPAECTVMGIPSISTNLSGFGCFMEEHIADPS AYGIYILDRFRSLDDSCSQLTSFLYSFCQQSRRQRIIQRNRTERLSDLLDWKYLGR YYMSARHMALAKAFPDHFTYEPHEVDATQGYRYPRAAVPPAPSLARHAAPHQ AEDEEPRDGPLGEDSERYDEEEAAKDRRNIRAPEWPRRASCSSSTGGSKRSNSV DTGPSSSLSTPTEPLSPTSSLGEERN
GYSd elc	MPLSRSLAVSSSLPGLEDWEDEFDPENAVLFEVAWEVANKVGGIYTVLQTKAKVT GDEWGDNYL VG PYTEQGVRTQVELLEPPTPELKRTLDSMNSKGCKVYFGRWLI EGGPLVLLDVGASAWALERWKGELWDTNIGVPWYDREANDAVLFGFLTTFW LGEFLAQNEEKPYVVAHFHEWLAGVGLCLCRARRLPVATIFTTHATLLGRYLCA GAVDFYNNLENFNVDKEAGERQIYHRYCMERAAAHCAHVFTTVSQITAIEAQHL LKRKPDIVTPNGLNVKKFSAMHEFQNLHAQSKARIQEFVRGHFYGHLDNFNLDKTL YFFIAGRYEFSNKGADVLEALARLNLLRVNGSEQTVVAFIMPARTNNFNVT LKGQAVRKQLWDTANTVKEKFGRKLYESLLVGSPLDMNKMLDKEDFTMMKRAI FATQRQSFPPVCTHNMLDDSSDPILTIRRIGLFNSSADRVKVFHFPEFLSSTSPLLP VDYEEFVRGCHLGVFPSYEPWGYTPAECTVMGIPSISTNLSGFGCFMEEHIADPS AYGIYILDRFRSLDDSCSQLTSFLYSFCQQSRRQRIIQRNRTERLSDLLDWKYLGR YYMSARHMALAKAFPDHFTYEPHEVDATQALGEERN
UGP	MSQDGASQFQEVILQELESVKKELEKILTTAASHEFEHTKKDLDGFRKLFHRFLQ EKGPSVDWGKIQRPPEDSIQPYEKIKARGLPDNISSVLNKL VVVKLNGGLGTSMG CKGPKSLIGVRNENTFLDLTVQQIEHLNKTNTDVPLVLMNSFNTDEDTKKILQK YNHCRVKIYTFNQSRYPINKESLLPIAKDVSYSGENTEAWYPPGHGDIYASFYNS GLLDTFIEEGKEYIFVSNIDNLGATVDLYILNHLMNPPNGKRCEFMVEVTNKTRAD VKGGTLTQYEGKLRLVEIAQVPKAHVDEFKSVSKFKIFNTNNLWISLGA VKRLQE QNAIDMEIIVNPKTLDGGLNVIQLETA VGAAIKSFENSLGINVPRSRFLPVKTTSDL LLVMSNLYSLNAGSLTMSEKREFPTVPLVKLGSSFTKVQDYLRRFESIPDMLELDH LTVSGDVTFGKNVSLKGTVIIIANHGDRIDIPPGAVLENKIVSGNLRILDH
GBE	MAAPAAPAAGETGPDARLEAALADVPELARLLEIDPYLKPFAADFQRRYKKFSQ VLHDIGENEGGIDKFSRGYESFGIHRCSGGIYCKEWAPGAEGVFLTGEFSGWNPF SHPYKKLEYGKWELYIPPKQNKSPLIHGSKLKVVITSKSGEILYRISPWAKYVVR ENNNVNYDWIHWAPEDPYKFKHSRPPKPRSLRIYESHVGISSHEGKIASYKHFTSN VLPRIKDLGYNCIQLMAIMEHAYYASFGYQITSFFAASSRYGTPEELKELVDTAHS MGIVVLLDVVHSHASKNSEDGLNMFDTGTDSCYFHS GPRGTHDLWDSRLFIYSSW EVLRFLLSNIRWWLEEYCFDGRFDGVTSMLYHHHGMGQGFSGDYNEYFGLQV DEDALIYMLANHLAHTLYPDSITIAEDVSGMPALCSPTSQGGGGFDYRLAMAIP DKWIQLLKEFKDEDWNMGNIVYTLTNNRYLEKCVAYAESHQALVGDKTAFW LMDAEMYTNMSVLAPFTPVIDRGIQLHKMIRLITHGLGGEGYLNFMGNEFGHPE WLD FPRKGNNESYHYARRQFNLTDDLLRYKFLNNFDRDMNRLEERCWLSAP QAYVSEKHEANKTITFERAGLLFIFNFHPSKSYTDYRVGTATPGKFKIVLDSDAAE YGGHQRLDHNTNYFAEAFEHNGRPYSLLVYIPSRVALILQNVDLQN

Table 1 (continued)

GYG	MTDQAFVTLTTNDAYAKGALVLGSSLKQHRTTRRMVVLTSPQVSDSMRKVLET VFDDVIMVDVLD SGDSAHLTLMKRPELGITLTKLHCWSLTQYSKCVFMDADTLV LSNIDDLFEREELSAAPDPGWPCFNSGVFVYQPSIETYNQLLHLASEQGSFDGGD QGLLNTYFSGWATTDITKHLPFVYNLSSISIYSYLP AFKAFGKNAKV VHFLGR TKP WNYTYNPQTKSVNCDSQDPTVSHPEFLNLWWD TFTTNVLP LLQHHGLVKDASSY LMMLSDLVYT LAFSCGFCRKEHVSGALSDLSFGEAPAAPQPSMSSEERKERWEQ GQADYMGADSFDNIKRKLD TYLQ
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Table 1 (continued)

Availability of data and materials

The dataset generated by this study is available in GSE281974 [<https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE281974>]. Additional data that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgements

We thank the team of Du Yanan at Tsinghua University for sharing the plasmid encoding Akaluc. We also thank Prof. George Guo-Qiang Chen at Tsinghua University for his advice on this project.

The authors declare that they have not used Artificial Intelligence in this study.

Additional information

Authors' contributions

Y.Y.X. and M.R. designed the experiments. Q.W. supervised the project. Y.Y. X., M. R., Z.Y. W., H.W. X. and B.Z. performed the experiments. H.Q. X., L. W., and S. C. analyzed the data. Q.W., Y.Y.X., M. R. and X.D. S. wrote the manuscript.

Ethical considerations

All animal experiments received approval from the institutional animal care and use committee of Tsinghua University (Project title: Study on different factors on the survivability of implanted MSCs in murine pulmonary fibrosis model, Approval number: 23-WQ1.G23-1, Date of approval: 2023-01-18).

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Reviewer #1 (Public review):

Summary:

This study provides the first evidence that glucose availability, previously shown to support cell survival in other models, is also a key determinant for post-implantation MSC survival in the specific context of pulmonary fibrosis. To address glucose depletion in this context, the authors propose an original, elegant, and rational strategy: enhancing intracellular glycogen stores to provide transplanted MSCs with an internal energy reserve. This approach aims to prolong their viability and therapeutic functionality after implantation.

Strengths:

The efficacy of this metabolic engineering strategy is robustly demonstrated both in vitro and in an orthotopic mouse model of pulmonary fibrosis.

<https://doi.org/10.7554/eLife.106023.2.sa2>

Reviewer #2 (Public review):

Summary:

In this article, the authors investigate enhancing the therapeutic and regenerative properties of mesenchymal stem cells (MSCs) through genetic modification, specifically by overexpressing genes involved in the glycogen synthesis pathway. By creating a non-phosphorylatable mutant form of glycogen synthase (GYSmut), the authors successfully increased glycogen accumulation in MSCs, leading to significantly improved cell survival under starvation conditions. The study highlights the potential of glycogen engineering to improve MSC function, especially in inflammatory or energy-deficient environments. However, critical gaps in the study's design, including the lack of validation of key findings,

limited differentiation assessments, and missing data on MSC-GYSmut resistance to reactive oxygen species (ROS), necessitate further exploration.

Strengths:

(1) **Novel Approach:** The study introduces an innovative method of enhancing MSC function by manipulating glycogen metabolism.

(2) **Increased Glycogen Storage:** The genetic modification of GYS1, resulting in GYSmut, significantly increased glycogen accumulation, leading to improved MSC survival under starvation, which has strong implications for enhancing MSC therapeutic properties in energy-deficient environments.

(3) **Potential Therapeutic Impact:** The findings suggest significant therapeutic potential for MSCs in conditions that require improved survival, persistence, and immunomodulation, especially in inflammatory or energy-limited settings.

(4) **In Vivo Validation:** The in vivo murine model of pulmonary fibrosis demonstrated the improved survival and persistence of MSC-GYSmut, supporting the translational potential of the approach.

Weaknesses:

(1) **Lack of Differentiation Assessments:** The study did not evaluate key MSC differentiation pathways, including chondrogenic and osteogenic differentiation. The absence of analysis of classical MSC surface markers and multipotency limits the understanding of the full potential of MSC-GYSmut.

(2) **Missing Validation of RNA Sequencing Data:** Although RNA sequencing data revealed promising transcriptomic changes in chondrogenesis and metabolic pathways, these findings were not experimentally validated, limiting confidence.

(3) **Lack of ROS Resistance Analysis:** Resistance to reactive oxygen species (ROS), an important feature for MSCs under regenerative conditions, was not assessed, leaving out a critical aspect of MSC function.

(4) **Limited Exploration of Immunosuppressive Properties:** The study did not address the immunosuppressive functions of MSC-GYSmut, which are critical for MSC-based therapies in clinical settings.

Conclusion:

The study presents an exciting new direction for enhancing MSC function through glycogen metabolism engineering. While the results show promise, key experiments and validations are missing, and several areas, such as differentiation capacity, ROS resistance, and immunosuppressive properties, require further investigation. Addressing these gaps would solidify the conclusions and strengthen the potential clinical applications of MSC-GYSmut in regenerative medicine.

<https://doi.org/10.7554/eLife.106023.2.sa1>

Author response:

The following is the authors' response to the original reviews.

| **Reviewer #1 (Public Review) :**

(1) Glycogen biosynthesis typically involves several enzymes. In this context, could the authors comment on the effect of overexpressing a single enzyme - especially a mutant version - on the structure or quality of the glycogen synthesized?

While quantitative molecular weight analysis of synthesized glycogen was not performed, we documented changes in glycogen particle morphology. GYSmut overexpression resulted in significantly enlarged singular glycogen granules, suggesting potential high molecular mass, while GYS-GYG co-overexpression in MSCs (GYG being the essential enzyme for glycogen synthesis initiation) produced a diffuse glycogen distribution pattern rather than particulate structures. We have incorporated this result as new Figure S2C.

These results suggest that overexpression of specific glycogen-metabolizing enzymes significantly influences glycogen structure. Consequently, targeted modulation of glycogen architecture and properties through key enzymes represents a potential avenue for future investigation.

(2) Regarding the in vitro starvation experiments (Figure 2C), what oxygen conditions (pO_2) were used? Are these conditions physiologically relevant and representative of the in vivo lung microenvironment?

Our in vitro starvation experiments (Figure 3C) were conducted under normoxic (21%). The oxygen concentration in human lungs is physiologically lower than atmospheric levels, with healthy individuals exhaling air containing approximately 16% oxygen (Thalakkotur Lazar Mathew, Diagnostics 2015). To our knowledge, direct measurements of alveolar oxygen concentration in pulmonary fibrosis are rare. Therefore, to evaluate the performance of GYSmut under hypoxic conditions, in the revised manuscript, Figure S2 has been augmented to include assessment of cell performance under combined hypoxia (oxygen concentration < 5%) and nutrient deprivation stress, which further corroborate the superiority of the GYSmut group over the control under different oxygen concentrations.

(3) In the in vitro model, how many hours does it take for the intracellular glycogen reserve to be completely depleted under starvation conditions?

While quantitative cell viability data were recorded up to 72 hours post-implantation (Fig 3C), we observed cell viability at approximately 96 hours. We noticed that the presence of glycogen particles exhibited a correlation with sustained cell viability. However, reliable quantitative assessment of glycogen became increasingly challenging upon significant depletion of viable cells, thereby limiting our measurements during later time points.

(4) For the in vivo model, is there a quantitative analysis of the survival kinetics of the transplanted cells over time for each group? This would help to better assess the role and duration of glycogen stores as an energy buffer after implantation.

We tracked the in vivo distribution and persistence of implanted MSCs using enzymatic activity quantification assays (using Gluc luciferase assay) and live animal imaging (using Akaluc luciferase). The revised manuscript includes quantitative analysis of the in vivo fluorescence imaging data, which has been supplemented as Figure S4. Glycogen-engineered MSCs and control cells were quantitatively assessed at three discrete time points post-implantation. This quantification revealed a transient divergence in cell viability between the experimental and control groups around day 7. However, fluorescence in both cohorts subsequently declined to similar levels over the extended observation period.

(5) Finally, the study was performed in male mice only. Could sex differences exist in the efficacy or metabolism of the engineered MSCs? It would be helpful to discuss whether

the approach could be expected to be similarly effective in female subjects.

We appreciate the reviewer's important question regarding potential sex differences. Our study used male mice based on three key considerations: 1) Clinical Relevance: Idiopathic pulmonary fibrosis (IPF) shows significant male predominance, with diagnosis rates 3.5-fold higher in men (37.8% vs 10.6%, $p < 0.0001$) and greater diagnostic confidence (Assayag et al., Thorax 2020). 2) Model Consistency: The bleomycin model (our chosen method) demonstrates more consistent fibrotic responses in male mice (Gul et al., BMC Pulm Med 2023). 3) Biological Rationale:

Estrogen's protective effects in females may confound therapeutic assessments (cited in Assayag et al.).

We fully acknowledge this limitation and will include female subjects in subsequent translational studies. The therapeutic principle should theoretically apply to both sexes, but we agree this requires experimental validation.

(6) The number of mice for each group and time point should be specified.

The manuscript text has been revised to enhance clarity, and the number of mice for each group and time point has been specified (line 170 to 182).

Reviewer #2 (Public Review):

(4) Inconsistencies in In Vivo Data: There is a discrepancy between the number of animals shown in the figures and the graph (three individuals vs. five animals), as well as missing details on how luciferase signal intensity was quantified, requiring further clarification.

To assess MSC survival in vivo, we employed two strategies utilizing distinct luciferases optimized for specific detection modalities. MSC viability was quantified ex vivo through Gaussia luciferase (Gluc) activity, leveraging its high sensitivity and established commercial assay kits ($n = 3$ mice per group per time point). For non-invasive longitudinal tracking within living animals, MSC distribution and viability were monitored via in vivo bioluminescence imaging using Akaluc luciferase, selected for its superior tissue penetration and sensitivity in situ ($n = 5$ mice per group). The manuscript text has been revised to enhance clarity, and the experiment protocols for luciferase signal detection and quantification has been added into Methods.

We fully agree that further investigation into the functional consequences of glycogen engineering in MSCs – encompassing core cellular functions, immunomodulatory properties, and associated signaling pathways – is important to fully elucidate the underlying mechanisms. Cellular metabolism is intrinsically intertwined with diverse physiological processes. Consequently, we believe that glycogen engineering exerts multifaceted effects on MSCs, likely extending beyond the modulation of any single specific pathway. Studying the metabolic perturbation induced by such engineering approaches in mammalian cells represents an interesting field. The exploration of these aspects remains an long-term research objective within our group.

Reviewer #2 (Recommendations for the authors):

(6) Clarification of Data in the Murine Model:

In Figure 4B, there is a discrepancy between the number of animals shown in the image (five) and those represented in the graph (three). This discrepancy needs clarification. Additionally, the study lacks information regarding the intensity of the signal in the

luciferase assays. It is unclear how luciferase expression in the mice was quantified, and providing this detail would enhance the understanding of the data presented.

We sincerely appreciate these valuable suggestions. We have revised the relevant text for greater clarity. Figure 4B and Figure 4C present results from two distinct experimental approaches, each employing different luciferase reporters and measurement methodologies, and different num of mice were used in these two experiments.

Quantitative data derived from the in vivo bioluminescence imaging has been supplemented as Figure S4. The experiment protocols for luciferase signal detection and quantification has been added into Methods.

To other recommendations of reviewer 2 :

We sincerely appreciate your valuable insights, which demonstrate your deep expertise. We fully agree that beyond nutrient availability, factors such as reactive oxygen species (ROS) and the immune microenvironment are also critical limitations affecting the survival and therapeutic efficacy of implanted MSCs.

We propose that glycogen engineering exerts broad effects on MSCs. These effects manifest as changes in multiple cellular characteristics, including proliferation, differentiation, surface marker expression, antioxidant capacity, and immunomodulatory activity – all crucial factors for the therapeutic purpose of MSCs.

We believe these changes likely involve complex networks of interconnected regulatory factors. The underlying mechanisms might be clarified through proteomic and metabolomic profiling.

However, comprehensively investigating these interconnected aspects requires significant time and resources. Some components of this research extend beyond the current scope of our project. Nevertheless, exploring these mechanisms remains an important objective, and we will actively work to investigate them further in our ongoing studies.

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